Cyber Security Fundamentals

Learning notes for document-loading and RAG practice

What is Cyber Security?

Cyber security is the practice of protecting computers, networks, applications, and data from unauthorized access, attacks, damage, or theft.

Common Threats

Common cyber threats include malware, phishing, ransomware, password attacks, denial-of-service attacks, and social engineering. Malware is harmful software. Ransomware may lock or encrypt files and demand payment.

Phishing

Phishing is a social-engineering technique in which an attacker sends a deceptive message to trick someone into revealing sensitive information or opening a harmful link or attachment. Users should verify the sender and avoid unexpected links.

Basic Protection

Use strong and unique passwords, enable multi-factor authentication, install security updates, back up important data, use trusted software, and be careful with links and attachments.

Encryption

Encryption transforms readable information into an encoded form using a cryptographic key. It helps protect confidentiality while data is stored or transmitted. Decryption converts the protected data back into readable form for an authorized party.

Incident Response

When an incident occurs, an organization should identify and contain the issue, preserve useful evidence, remove the cause, restore systems safely, and review what happened to reduce future risk.